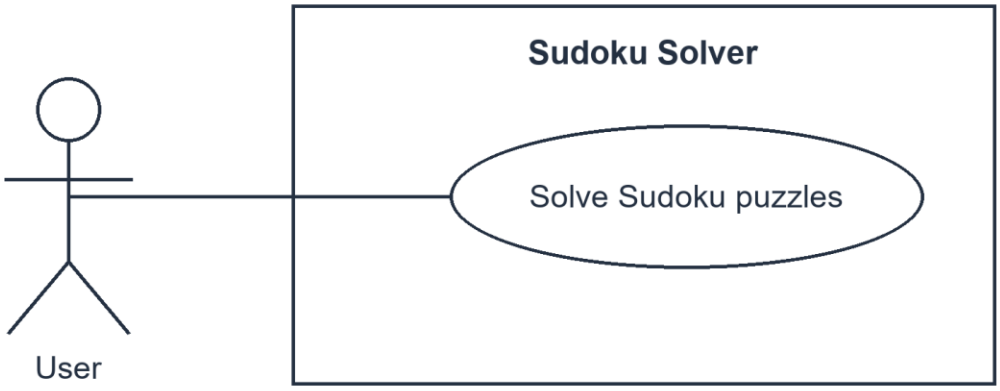


a. Identify the use case: Solve Sudoku puzzles. The actor is the **User**.

b. Brief description: This use case allows the user to run the program, view each original puzzle, and see all solutions found. If no filenames are provided, the program uses `puzzle1.txt` through `puzzle5.txt`.

c. Step-by-step flow:

1. The user runs the program.
2. The system opens a puzzle file and checks its contents.
3. The system displays the original puzzle.
4. The system checks the starting numbers and solves the puzzle.
5. The system displays each solution and the total number found.
6. The system repeats these steps for the remaining files.

d. Detailed description:

- **Precondition:** The Sudoku Solver program is available to run.
- **Trigger:** The user starts the program, optionally providing puzzle filenames.
- **Main flow:** The system opens the first file and reads its 81 cells. It displays the original puzzle, checks for duplicate starting numbers, and searches for solutions. It prints every solution and the total number found. It then processes the next file.
- **Alternate flows:** If a file cannot be opened or has an invalid format, the system prints an error and continues to the next file. If the starting numbers contain a duplicate, it prints “Invalid starting puzzle.” If a valid puzzle has no solution, it prints “No solution found.”
- **Postcondition:** The results or an error message have been displayed for each requested file.